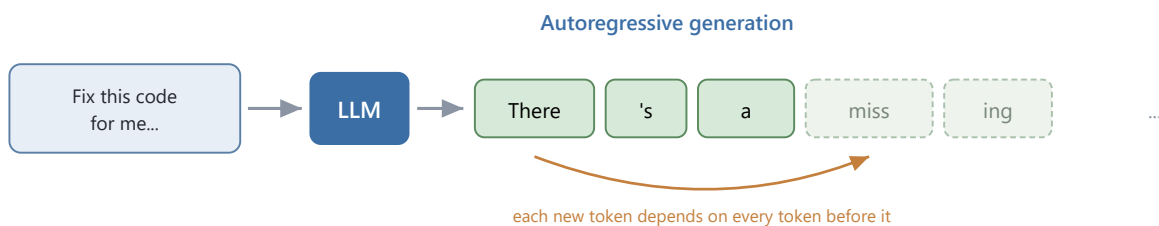


Part 1: How the Model Behaves

Before we look at a single matrix, let's watch what a model actually does when you send it a message. Everything in Part 1 is visible from the outside — no math required.

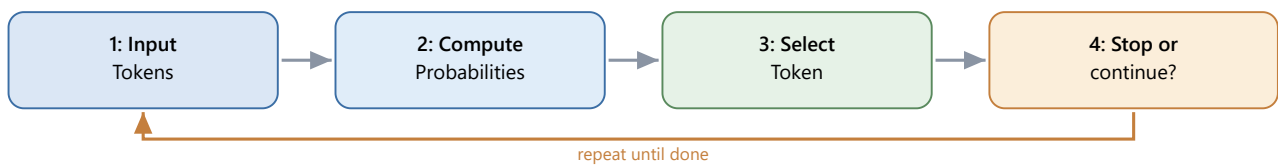
The Autoregressive Loop

You send: "Fix this code for me...". The model replies: "There's a missing colon...". It looks like the model wrote a sentence. It didn't. It wrote **one token**, then looked at everything again including that token, and wrote one more. And again. And again.



The model produces one token, appends it to its own input, and runs again.

That loop has exactly four steps, and we are going to spend the rest of Part 1 on them.



The whole of "generation" is these four steps in a circle.

Step 1: Input Tokens and the Context Window

The model does not see your screen, your files, or your history. It sees a flat list of tokens sitting in its **context window**. Whatever is in that window is the model's entire world for this one step.